



MANIPAL INSTITUTE OF TECHNOLOGY

BENGALURU

(A constituent unit of MAHE, Manipal)

Sl.No. **00244**

SESSIONAL RECORD

DEPT.: **COMPUTER SCIENCE**
ENGG

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SEMESTER: **VIII**

SECTION: **CSE-CORE-A**

ROLL NO.: **-**

SUBJECT: **BLOCKCHAIN TECHNOLOGY CSE-4033**

DATE OF TESTS: **11/10/2025**

MAX MARKS : 15/20

MARKS OF: **I TEST**

II TEST

MARKS FOR VALUATION - I TEST

MARKS FOR VALUATION - II TEST

Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS	Q. No.	FOR THE TEACHER TO AWARD MARKS					TOTAL MARKS
	a	b	c	d	e			a	b	c	d	e	
1						8	1						
2						5	2						
3						4	3						
4						5	4						
5						2	5						
6							6						
TOTAL IN FIGURES						24	TOTAL IN FIGURES						

IN WORDS _____

IN WORDS _____

Signature of the Candidate

Signature of the Teacher

BLOCK-CHAIN TECHNOLOGY

(MCQS QUESTION AND ANSWER)

- Q 1. ~~A~~ option (c) Run applications on a peer-to-peer (P2P).
- Q 2. option (c) By protecting data ~~that~~ that has been submitted
- Q 3. option (d) Public blockchain ~~offer~~ rewards for mining
- Q 4. option (c) Attacker gained control ~~of~~ more than 51% of network's computing
- Q 5. option (d) Miner node
- Q 6. option (a) certificate authority
- Q 7. option (b) stack based
- Q 8. option (a) Gas
- Q 9. option (c) Both A and B
- Q 10. option (a) A transaction and block verification protocol

Answer Key

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
C	C	D	C	D	A	B	A	C	A

Q 11) Analyze the need for consensus in blockchain systems. Discuss Proof-of-work (POW) and proof of Burn (POB) algorithms

A 11) consensus in blockchain systems

proof of work algorithm (POW)

Mechanisms: Miners compute to solve a complex mathematical puzzle. This involves repeatedly hash the data in a new block along with a random number until resulting hash number is below a certain bit, target value set by the network difficulty

Function: The First miner to find the solution gets to add their first block to the blockchain and is rewarded with the cryptocurrency. The 'work' is immense computational power spent to find their solution. The process is difficult and but resource-intensive to perform but very easy for other nodes to verify.

Pros - Highly secure and has been extensively tested.

cons. Enormous energy consumption, can lead to centralization of mining power and has relatively low transaction throughput

Proof of Burn (PoB): Here miners burn coins to provably unspendable address. By burning these coins they ~~prove~~ their commitment ~~towards~~ mining to the network and earn the right to mine new blocks.

The more coin the user burn the higher probability ~~they~~ ^{miners} will have to mine the next block.

Function: It acts as a long term investment in the network security instead of investing in physical mining hardware, users invest by sacrificing short term wealth

pros: more energy efficient than POW

cons: still 'waste' resource by destroying value; can favor early adopters who acquired coins at low price.

Q 12) compare Multisig, Pay to-script (P2SH), and Proof of - Burn transactions in terms of usability and security.

Feature	Multisig (Multisignature)	pay to script	proof of Burn
security	A type of digital signature that requires M-of-N keys to authorize transaction.	A transaction type where funds are sent to the hash of a script. The recipient must provide an original script to full fill condition to spend funds	A transaction that sends coins to a provably unspendable address, effectively destroying them.
purpose	To increase security by distributing control over funds used for joint accounts, corporate governance and crowd	To simplify transactions involving complex scripts. It makes a complex transaction look like standard one to sender	To demonstrate long-term commitment to a network, bootstrap a new cryptocurrency or earn mining privilege
usability	For spender: complex, requires coordination between multiple key holders For Sender: can be complex as they must handle log		

Q 13. Infer the Byzantine General problem and its role in blockchain consensus mechanisms.

A 13. Byzantine General problem and its role in blockchain consensus mechanisms:

The Byzantine General problem

- 1) It describes a scenario where a group of generals, each leading a part of an army, are surrounding to an enemy city. They must agree on a unified plan - either to attack or retreat. The challenge is to communicate via messengers and some of the generals may be traitors (Byzantine). A traitor might send an "attack" message to one general and a "retreat" message to another which can cause chaos. The problem is to find a protocol that allows all loyal generals to reach the same decision regardless of traitors action.

Analogy: In block-chain the "generals" are distributed nodes in the network. The "plan" is the current state of the ledger.

"Traitors" are malicious nodes that might try to cheat the system by double-spending creating invalid blocks.

Solution: The analogy described in the nodes of the network Blockchain consensus can be handled mechanisms are designed to be:-

Byzantine Fault Tolerance (BFT)

They are practical solutions to BFT. They provide a way for the network of nodes to agree on a single source of truth.

Example: Proof of work (POW) solves the BFT by making it economically unfeasible for "traitors" to disrupt the consensus. A

malicious nodes would need control over 51% of the entire network's computing power to reliably overpower the honest nodes and re-write the ledger. The cost is so high that it incentivises all participants to act honestly.

Q 14. Define atomic broadcast and discuss its importance in distributed systems.

A 14. Definition of Atomic Broadcast:

Atomic broadcast is a communication primitive in distributive systems that ensures messages are delivered reliably and in the same order to all participants. It guarantees two principles:

1. Total order: All nodes correct node deliver messages in the exact same sequence. If Node A process message M_1 then M_2 , Nodes B will process M_1 then M_2 never.